Factors, primes and divisibility

L2-NT1 Level 2 Intermediate · Number theory

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How to read divisors, HCFs, LCMs and factorials straight off a prime factorisation.
By the end you can count the divisors of a number, and the divisors with a given property, from its prime factorisation; find HCFs and LCMs by comparing exponents or by Euclid's subtraction idea; find the exact power of a prime that divides a factorial.
Prerequisites: Factors, multiples and primes (L1-NT1). **Contests:** IMC, Grey Kangaroo, Pink Kangaroo, Cayley, Hamilton, JMO, AMC 8, AMC 10. **Time:** about 75 minutes.

1 Lesson

How many positive whole numbers divide 3600? You could list pairs: 1 and 3600, 2 and 1800, 3 and 1200, and so on, but it is hard to be sure that no pair has been missed. Instead, write $3600 = 2^4 \times 3^2 \times 5^2$. Every divisor of 3600 is built by choosing how many 2s, how many 3s and how many 5s to take: between 0 and 4 twos, between 0 and 2 threes, and between 0 and 2 fives. That is $5 \times 3 \times 3 = 45$ choices, so 3600 has exactly 45 divisors.

The method of this unit is to factorise a number into primes and read off what we want from the exponents. Questions about divisors, highest common factors, lowest common multiples, perfect squares and factorials all reduce to statements about exponents.

1.1 Primes and prime factorisation

A prime is a positive integer with exactly two positive divisors, 1 and itself. So 1 is not prime (it has only one divisor), and 2 is the only even prime. The first few primes are 2, 3, 5, 7, 11, 13, 17, 19, 23, 29.

> **The Fundamental Theorem of Arithmetic.**
> Every integer $n > 1$ can be written as a product of primes, and in only one way apart from the order of the factors.

Existence is fairly trivial. If n is prime, stop. Otherwise $n = ab$ with $1 < a, b < n$, so split a and b in the same way. The numbers keep getting smaller, so the process must end. Uniqueness is the hard part, and it rests on one fact, *Euclid's lemma*: if a prime p divides a product ab, then p divides a or p divides b. (We take it on trust here; you will see why it is true when we meet modular inverses in Level 3.) Given two factorisations of the same number, each prime in the first divides the product in the second, so by the lemma it is one of the primes there. Cancel it from both sides and repeat.

To test whether n is prime you only need to try primes up to $\sqrt{n}$. If $n = ab$ with $a \le b$, then $a^2 \le ab = n$, so $a \le \sqrt{n}$: any factorisation has a factor no bigger than the square root. To check 211, try 2, 3, 5, 7, 11 and 13. None divides it, and $17^2 = 289 > 211$, so 211 is prime.

Setters often use the current year, so it is worth remembering its factorisation: $2024 = 2^3 \times 11 \times 23$,

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$2025 = 3^4 \times 5^2 = 45^2$, $2026 = 2 \times 1013$ with 1013 prime, and 2027 is prime.

1.2 Reading a number from its factorisation

Write $n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$. A positive integer d divides n exactly when $d = p_1^{b_1} p_2^{b_2} \cdots p_k^{b_k}$ with $0 \le b_i \le a_i$ for every i. In words: a divisor uses only the primes of n, and never more copies of a prime than n has. (If d involved a prime q not in the list, then q would divide n, which uniqueness forbids.)

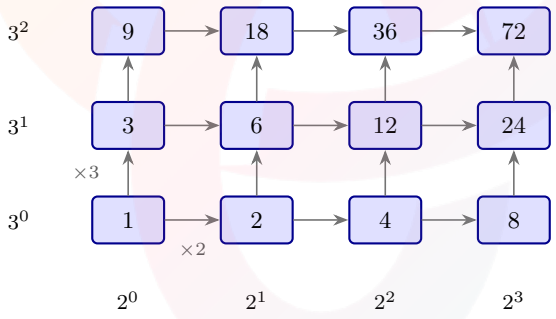
Counting and adding divisors.
If $n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$, then the number of positive divisors of n is

$$d(n) = (a_1 + 1)(a_2 + 1) \cdots (a_k + 1),$$

and the sum of the positive divisors is

$$\sigma(n) = (1 + p_1 + \cdots + p_1^{a_1})(1 + p_2 + \cdots + p_2^{a_2}) \cdots (1 + p_k + \cdots + p_k^{a_k}).$$

The count follows from the multiplication principle: there are $a_1 + 1$ choices for the exponent of p_1 (remember that 0 is a choice), $a_2 + 1$ for the exponent of p_2, and so on. For the sum, note that expanding the brackets produces one term for each choice of exponents, so each divisor appears once.



4 choices for the power of 2, 3 for the power of 3: $4 \times 3 = 12$ divisors

The picture shows this for $72 = 2^3 \times 3^2$. Put the divisors in a grid with the power of 2 going across and the power of 3 going up. There are 4 columns and 3 rows, so $d(72) = 12$.

Divisors also come in pairs: d goes with n/d. In each pair one member is below $\sqrt{n}$ and the other is above, except when $d = \sqrt{n}$, which pairs with itself. So n **has an odd number of divisors if and only if** n **is a perfect square**. The formula agrees: $(a_1 + 1) \cdots (a_k + 1)$ is odd only when every a_i is even.

To count divisors *with a property*, restrict the exponent choices. A divisor is odd when its exponent of 2 is 0. It is a multiple of 10 when its exponents of 2 and 5 are both at least 1. It is a perfect square when every exponent is even.

Example 1.
How many positive divisors of 5400 are multiples of 15?
Solution.
We factorise: $5400 = 54 \times 100 = (2 \times 3^3) \times (2^2 \times 5^2) = 2^3 \times 3^3 \times 5^2$. A divisor has the form

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$2^a3^b5^c$ with $0 \le a \le 3$, $0 \le b \le 3$ and $0 \le c \le 2$.
Being a multiple of $15 = 3 \times 5$ means $b \ge 1$ and $c \ge 1$. So a has 4 choices, b has 3 (namely 1, 2 or 3) and c has 2, giving $4 \times 3 \times 2 = 24$ divisors.
As a check, we count a second way. The divisors we want are the numbers $15k$ that divide 5400, which happens when k divides $5400/15 = 360$. Since $360 = 2^3 \times 3^2 \times 5$ has $4 \times 3 \times 2 = 24$ divisors, the two answers agree.

Example 2.
What is the smallest positive integer with exactly 12 positive divisors?

Solution.
Testing numbers one at a time would work here, but not for a question asking for 48 divisors, so we work from the formula. We need $(a_1+1)(a_2+1)\cdots = 12$, so first list the ways to write 12 as a product of factors that are each at least 2: 12, 6×2, 4×3 and $3 \times 2 \times 2$. These give the exponent patterns (11), $(5,1)$, $(3,2)$ and $(2,1,1)$.
For each pattern the smallest number puts the biggest exponent on the smallest prime. That gives $2^{11} = 2048$, $2^5 \times 3 = 96$, $2^3 \times 3^2 = 72$ and $2^2 \times 3 \times 5 = 60$. The smallest is 60.
Spreading the exponents over several small primes usually gives the smallest number, since each extra factor of 2 doubles it. This is a rule of thumb, not a theorem, so always compare all the candidates.

1.3 HCF and LCM

HCF and LCM from factorisations.
If $a = \prod p^{\alpha_p}$ and $b = \prod p^{\beta_p}$ (allowing exponent 0), then

$$\operatorname{hcf}(a,b) = \prod p^{\min(\alpha_p,\beta_p)}, \qquad \operatorname{lcm}(a,b) = \prod p^{\max(\alpha_p,\beta_p)}.$$

Since $\min(x,y) + \max(x,y) = x + y$, it follows that $\operatorname{hcf}(a,b) \times \operatorname{lcm}(a,b) = ab$.

These follow from the previous section. A common divisor may use each prime no more often than *both* numbers do, and a common multiple must use each prime at least as often as *either* does. For example, $360 = 2^3 \times 3^2 \times 5$ and $84 = 2^2 \times 3 \times 7$, so $\operatorname{hcf}(360,84) = 2^2 \times 3 = 12$ and $\operatorname{lcm}(360,84) = 2^3 \times 3^2 \times 5 \times 7 = 2520$. Check: $12 \times 2520 = 30240 = 360 \times 84$.

When the numbers are large, factorising can be slow, and we use Euclid's idea instead: **any common divisor of a and b also divides $a - kb$ for every integer k**, and any common divisor of $a - kb$ and b divides a. So $\operatorname{hcf}(a,b) = \operatorname{hcf}(a-kb,b)$, and you can keep replacing the bigger number by its remainder on division by the smaller:

$$\operatorname{hcf}(1001,221) = \operatorname{hcf}(117,221) = \operatorname{hcf}(117,104) = \operatorname{hcf}(13,104) = 13.$$

The same idea handles remainder questions. If dividing a by n leaves remainder r, then n divides $a - r$. So if two numbers leave known remainders on division by the same n, subtract the remainders and look for common divisors of what is left.

Example 3.
How many ordered pairs (a,b) of positive integers have $\operatorname{hcf}(a,b) = 12$ and $\operatorname{lcm}(a,b) = 360$?

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Solution.
One might use $ab = 12 \times 360 = 4320$ and count the factor pairs of 4320. That overcounts: $48 \times 90 = 4320$, but $\text{hcf}(48, 90) = 6$. The product alone does not pin down the HCF.
Instead, take out the HCF. Write $a = 12x$ and $b = 12y$. Then x and y have no common factor bigger than 1, since otherwise the HCF of a and b would be bigger than 12. For two such numbers the LCM is just the product, so $\text{lcm}(a, b) = 12xy$, and the condition becomes $xy = 30$.
So we need ordered pairs of positive integers with product $30 = 2 \times 3 \times 5$ and no common factor. That means each prime goes entirely to x or entirely to y. Each of the three primes has 2 choices, so there are $2^3 = 8$ ordered pairs. They are $(a, b) = (12, 360), (24, 180), (36, 120), (60, 72)$ and their reverses, and the answer is 8.

1.4 Primes inside factorials

We often need the factorisation of a factorial without multiplying it out. How many times does 2 divide 12!? Every even number up to 12 gives one factor of 2, which is $\lfloor 12/2 \rfloor = 6$ factors. The multiples of 4 each give a second one, which is $\lfloor 12/4 \rfloor = 3$ more. The multiples of 8 give a third, which is 1 more, and there are no multiples of 16. The total is $6 + 3 + 1 = 10$.

Legendre's formula.
The exponent of the prime p in $n!$ is

$$\left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \left\lfloor \frac{n}{p^3} \right\rfloor + \cdots,$$

where the terms become 0 once $p^k > n$.

Here $\lfloor x \rfloor$ means x rounded down. A number divisible by p^j but not by p^{j+1} is counted once in each of the first j terms, so j times in all, which is exactly how many factors of p it contributes. In particular, the number of zeros at the end of $n!$ is the exponent of 5, because there are always more 2s than 5s. So 100! ends in $20 + 4 = 24$ zeros.

Example 4.
How many positive divisors of 12! are perfect squares, and what is the largest of them?

Solution.
First we need 12! as a product of primes, and Legendre's formula gives the exponents directly: for 2 it is $6 + 3 + 1 = 10$, for 3 it is $4 + 1 = 5$, for 5 it is 2, and for 7 and 11 it is 1. So $12! = 2^{10} \times 3^5 \times 5^2 \times 7 \times 11$.
A square divisor needs every exponent even. The exponent of 2 can be 0, 2, 4, 6, 8 or 10 (6 choices), the exponent of 3 can be 0, 2 or 4 (3 choices), the exponent of 5 can be 0 or 2 (2 choices), and the exponents of 7 and 11 must be 0. That gives $6 \times 3 \times 2 = 36$ square divisors.
The largest takes the biggest even exponent each time: $2^{10} \times 3^4 \times 5^2 = (2^5 \times 3^2 \times 5)^2 = 1440^2$.
It is good practice to test the method on a small case. It predicts that $12 = 2^2 \times 3$ has $2 \times 1 = 2$ square divisors, and indeed they are 1 and 4.

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Common traps.

- 1 is not prime, but it divides everything. If a question says "factors other than 1 and the number itself", subtract 2 from the count.
- The number of divisors uses $a_i + 1$, not a_i. Forgetting that 0 is an allowed exponent is the commonest slip of all.
- Divisible by 4 and by 6 means divisible by $\text{lcm}(4, 6) = 12$, not by 24. You may multiply the two numbers only when they have no common factor.
- $\text{hcf} \times \text{lcm} = ab$ is true for two numbers, not three: $\text{hcf}(2, 2, 2) \times \text{lcm}(2, 2, 2) = 4$, not 8.
- Knowing ab does not tell you $\text{hcf}(a, b)$. Pull out the HCF first, as in Example 3.

1.5 How to spot this in a contest

The words *factor*, *divisor*, *multiple*, *prime* and *perfect square* are the obvious signals, but the same ideas appear in word problems. Things that repeat and coincide (bells, buses, flashing lights, meshing gears) are LCM questions. Cutting a rectangle into equal squares, or sharing things into equal groups with nothing left over, are HCF questions. "The product is a perfect square" means check whether every exponent is even, and "a perfect cube" means every exponent is a multiple of 3. Trailing zeros, or "the largest power of p that divides", mean Legendre. Remainders from the same divisor mean subtract and take a common factor.

Whenever a large specific number appears, especially the year, factorise it first. If you are listing more than about ten things, check whether the count can be read off from exponents instead.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ Call a positive integer *thrifty* if it has exactly three positive divisors. What is the sum of all thrifty numbers less than 200?

Problem 2 ⋆⋆ How many positive divisors of 7200 are multiples of 6 but not multiples of 12?

Problem 3 ⋆⋆ For how many positive integers n is $\text{lcm}(12, 18, n) = 720$?

Problem 4 ⋆⋆⋆ As n runs through the positive integers, what is the largest possible value of $\text{hcf}(n^2 + 5,\ n + 2)$?

Problem 5 ⋆⋆⋆ A positive integer n has exactly 12 positive divisors. The number $2n$ has exactly 15 positive divisors, and $3n$ has exactly 16. Find n.

Problem 6 ⋆⋆⋆⋆ For how many positive integers $n \leq 1000$ does n have an odd number of odd positive divisors?

3 Contest ladder

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Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **UKMT IMC 2016 Question 16** ⋆
   Compare each exponent with the exponents in 24.
2. **AMC 8 2022 Problem 3** ⋆
   Factorise 100 and list systematically, smallest factor first.
3. **UKMT IMC 2021 Question 6** ⋆
   Factorise 240 before you start listing.
4. **AMC 8 2018 Problem 18** ⋆⋆
   Prime factorisation, then the divisor-counting formula.
5. **AMC 10A 2022 Problem 7** ⋆⋆
   Work prime by prime with minimum and maximum exponents.
6. **AMC 10A 2023 Problem 5** ⋆⋆
   Collect the 2s and the 5s into powers of 10.
7. **UKMT IMC 2019 Question 24** ⋆⋆⋆
   Subtract the remainders and look for a common factor.
8. **AMC 10A 2019 Problem 11** ⋆⋆⋆
   Restrict the exponents, and do not count the overlap twice.
9. **AMC 10B 2017 Problem 20** ⋆⋆⋆
   Legendre's formula for the prime 2.
10. **AMC 10B 2021 Problem 12** ⋆⋆⋆
    Use the product formula for the sum of divisors and separate off the power of 2.
11. **UKMT IMC 2023 Question 23** ⋆⋆⋆
    Track the exponent of each prime in the product of all nine digits.
12. **AMC 10A 2018 Problem 22** ⋆⋆⋆⋆
    Write each condition prime by prime and see what the gcds force.

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4 Hints

1. Use the divisor-counting formula. Which exponent patterns give a product of brackets equal to 3?
2. Factorise 7200, then work out what being a multiple of 6 but not of 12 says about the exponent of 2 and the exponent of 3.
3. Work one prime at a time. For each prime, the largest of the three exponents must match the exponent in 720.
4. A common divisor of two numbers divides any combination of them. Which multiple of $n+2$ is very close to n^2+5?
5. Write $n = 2^a 3^b m$ where m has no factor 2 or 3, and compare the number of divisors of n with that of $2n$ as a ratio.
6. The odd divisors of n are exactly the divisors of the largest odd number dividing n. When does a number have an odd number of divisors?

5 Solutions

Problem 1

Call a positive integer thrifty *if it has exactly three positive divisors. What is the sum of all thrifty numbers less than 200?*

If $n = p_1^{a_1} \cdots p_k^{a_k}$, then n has $(a_1+1)\cdots(a_k+1)$ divisors. For this product to equal 3, which is prime, there must be just one prime, with exponent 2, so $n = p^2$ for a prime p. The squares of primes below 200 are 4, 9, 25, 49, 121 and 169 (the next one, $17^2 = 289$, is too big). Their sum is 377.

Answer: 377

Problem 2

How many positive divisors of 7200 are multiples of 6 but not multiples of 12?

We have $7200 = 72 \times 100 = 2^5 \times 3^2 \times 5^2$, so a divisor is $2^a 3^b 5^c$ with $0 \le a \le 5$, $0 \le b \le 2$ and $0 \le c \le 2$. Being a multiple of 6 means $a \ge 1$ and $b \ge 1$. Being a multiple of 12 as well would mean $a \ge 2$ and $b \ge 1$. So the divisors we want have $b \ge 1$ and $a = 1$ exactly. Then a has 1 choice, b has 2 choices (1 or 2) and c has 3 choices, giving $1 \times 2 \times 3 = 6$ divisors.

Answer: 6

Problem 3

For how many positive integers n is $\operatorname{lcm}(12, 18, n) = 720$*?*

We have $12 = 2^2 \times 3$, $18 = 2 \times 3^2$ and $720 = 2^4 \times 3^2 \times 5$. Since n divides the LCM, $n = 2^a 3^b 5^c$ with $a \le 4$, $b \le 2$, $c \le 1$. The LCM takes the largest exponent of each prime. For the prime 2, the largest of $2, 1, a$ must be 4, which forces $a = 4$. For the prime 3, the largest of $1, 2, b$ must be 2,

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which holds for $b = 0, 1$ or 2. For the prime 5, the largest of $0, 0, c$ must be 1, which forces $c = 1$. So n is 80, 240 or 720, and there are 3 such n.

Answer: 3

Problem 4

As n runs through the positive integers, what is the largest possible value of $\text{hcf}(n^2 + 5,\ n + 2)$*?*

Let $g = \text{hcf}(n^2 + 5, n + 2)$. Since g divides $n + 2$, it divides $(n + 2)(n - 2) = n^2 - 4$. It also divides $n^2 + 5$, so it divides the difference $(n^2 + 5) - (n^2 - 4) = 9$. Hence $g \leq 9$ for every n. The value 9 is achieved: when $n = 7$ we get $n + 2 = 9$ and $n^2 + 5 = 54 = 6 \times 9$, whose HCF is 9. So the largest possible value is 9.

Answer: 9

Problem 5

A positive integer n has exactly 12 positive divisors. The number $2n$ has exactly 15 positive divisors, and $3n$ has exactly 16. Find n.

Write $n = 2^a 3^b m$, where m is not divisible by 2 or 3. Then n has $(a + 1)(b + 1)d(m)$ divisors, $2n$ has $(a + 2)(b + 1)d(m)$, and $3n$ has $(a + 1)(b + 2)d(m)$, where $d(m)$ is the number of divisors of m. Dividing, $\frac{a+2}{a+1} = \frac{15}{12} = \frac{5}{4}$, so $4a + 8 = 5a + 5$ and $a = 3$. Similarly $\frac{b+2}{b+1} = \frac{16}{12} = \frac{4}{3}$, so $3b + 6 = 4b + 4$ and $b = 2$. Then $12 = 4 \times 3 \times d(m)$ gives $d(m) = 1$, so $m = 1$ and $n = 2^3 \times 3^2 = 72$. As a check, $144 = 2^4 \times 3^2$ has $5 \times 3 = 15$ divisors and $216 = 2^3 \times 3^3$ has $4 \times 4 = 16$.

Answer: 72

Problem 6

For how many positive integers $n \leq 1000$ does n have an odd number of odd positive divisors?

Write $n = 2^k m$ with m odd. The odd divisors of n are exactly the divisors of m, and a number has an odd number of divisors exactly when it is a perfect square. So we need $m = s^2$ for some odd s, that is, $n = 2^k s^2$.

If k is even, then $n = (2^{k/2} s)^2$ is a perfect square. If k is odd, then $n = 2 \times (2^{(k-1)/2} s)^2$ is twice a perfect square. Conversely, if n is a square or twice a square, every odd prime appears in n to an even power, so its odd part m is a square. Hence the numbers we want are exactly the squares and the doubles of squares. No number is both, since the exponent of 2 is even in a square and odd in twice a square.

There are 31 squares up to 1000 (as $31^2 = 961$ and $32^2 = 1024$), and the doubles of squares $2t^2 \leq 1000$ need $t^2 \leq 500$, so $t \leq 22$ (as $22^2 = 484$ and $23^2 = 529$), giving 22 more. The answer is $31 + 22 = 53$.

Answer: 53